



# JAIPUR NATIONAL UNIVERSITY

## Semester End Examination Dec.2024

|                             |                                                                                                                                                                                                                                                                                                  |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name of School:             | School of Engineering and Technology                                                                                                                                                                                                                                                             |
| Name of Programme / Course: | B.Tech (AIML) / B.Tech (Biomedical) /<br>B.Tech (Biotechnology) / B.Tech (Chemical) /<br>B.Tech (Civil Engineering) / B.Tech<br>(Computer Science) / B.Tech (Cyber<br>Security) / B.Tech (Data Science) / B.Tech<br>(Electrical Engineering) / B.Tech (Food<br>Technology) / B.Tech (Mechanical) |
| Semester/Year:              | I                                                                                                                                                                                                                                                                                                |
| Paper Code:                 | BTEEEDS101T22                                                                                                                                                                                                                                                                                    |
| Title of Paper:             | Fundamental of Electrical and Electronics                                                                                                                                                                                                                                                        |
| Time: 3 Hours               | Max. Marks: 60                                                                                                                                                                                                                                                                                   |

### INSTRUCTIONS / NOTES FOR THE EXAMINEE

Attempt any five Questions from section A and attempt any four Questions from section B.

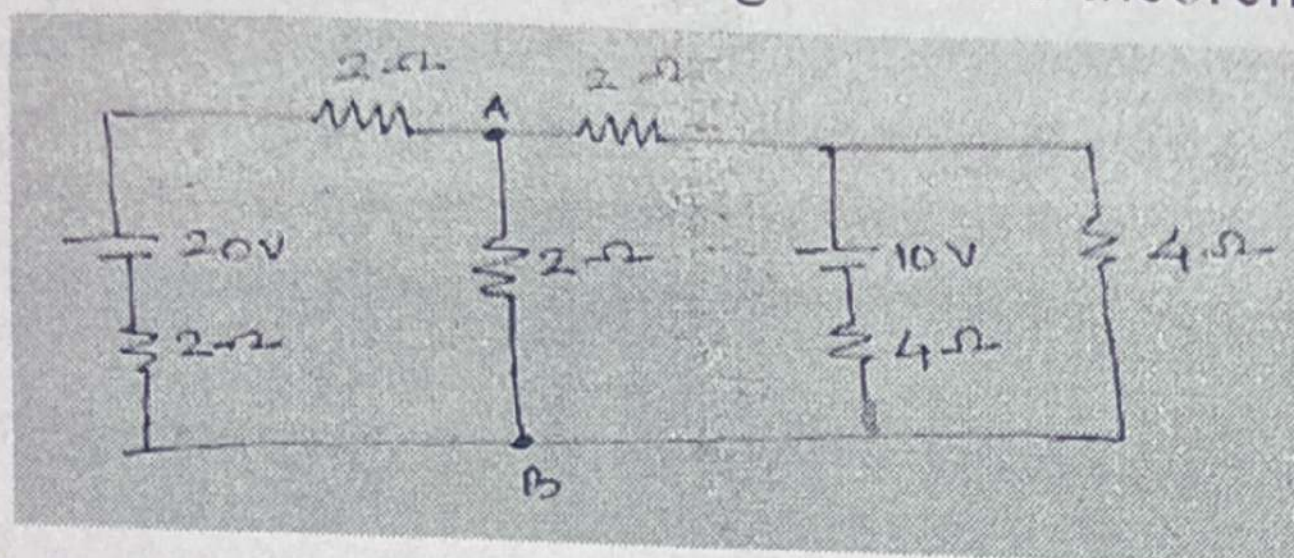
| Question Number | SECTION-A                                                                                                                                                                                                      | Marks Allotted |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <del>Q.1</del>  | 2 Explain the electromotive and magneto motive force.                                                                                                                                                          | 4              |
| Q.2             | State and explain the Faraday's Law of electromagnetic induction.                                                                                                                                              | 4              |
| <del>Q.3</del>  | 2 Define the following terms: <ul style="list-style-type: none"><li>Root Mean Square value of Alternating Supply</li><li>Average Value of Alternating Supply</li><li>Form Factor</li><li>Peak Factor</li></ul> | 4              |
| <del>Q.4</del>  | 2 Explain the modulation and demodulation.                                                                                                                                                                     | 4              |
| Q.5             | Explain Photovoltaic cell.                                                                                                                                                                                     | 4              |
| <del>Q.6</del>  | Explain the load cell                                                                                                                                                                                          | 4              |

Q.7 Explain the L-C filter

### SECTION-B

~~Q.1~~<sup>8</sup> Deduce an expression for RLC series circuit with the help of suitable circuit arrangement. 10

Q.2 Find current  $I$  at terminal AB using Norton's theorem. 10



~~Q.3~~<sup>8</sup> Derive EMF equation of I-phase transformer. 10

Q.4 Draw and explain V-I characteristic of PN junction diode. 10

~~Q.5~~ Draw and explain bipolar junction transistor. 10

~~Q.6~~ Write short note on thermocouple and RTD. 10